



Contact

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Address

Bassano del Grappa, VI
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Italy

Skills

Languages

Italian (Mother tongue),
English (Fluent)

Lab

EEG-based MI Decoding,
BCI for Human-Robot
Interaction, Experimental
Design & Data Analysis

Software

Linux, LaTeX, ROS

Programming

C++, Matlab, Python3

Job related

Project Management &
Critical Thinking,
Collaborative & Organized,
Creative Problem-Solving,

Personal

Cooking, Hiking, Wrestling

Piero Simonetto

Control System Engineer

About Me:

Ph.D. student specializing in Human-Robot Interaction (HRI) and Brain-Computer Interfaces (BCI). My research focuses on integrating human brain activity with robotic systems to enhance interaction and autonomy. Passionate about teamwork, cutting-edge technology, and applied research in assistive robotics.

Education:

02/2023 - present, University of Padua Ph.D.: Information Engineering

Thesis: "Human-Machine Interfaces for Industry 5.0: Advancing Human-Robot Interaction through Real-Time Decoding of User Cognitive States."

09/2020 - 10/2022, University of Padua LM-25: Control System Engineering

Thesis: "Fault detection for boolean systems"

10/2017 - 09/2020, University of Padua L-8: Ingegneria dell'Informazione

Thesis: "Classical and Quantum entropy"

09/2012 - 07/2017, I.T.I.S. E. Fermi High school on Computer Science

Publications:

P. Simonetto, A. Polato, F. Pasti, M. R. ur Rahman, S. Vascon, N. Bellotto, E. Menegatti, S. Tortora, and L. Tonin "A Multi-Layer Navigation System for Smart Wheelchairs with a Limited Input Interface" - 2026 *IEEE Intelligent Robots & System (IEEE IROS)*

P. Simonetto, S. Toniolo, S. Tortora, E. Menegatti, L. Tonin "Enhancing Motor Imagery Decoding with Environmental Context During Robot Control" - 2025 *System Man & Cybernetics Conference (SMC)*

D. De Lazzari, P. Simonetto, N. Turcato, L. Tonin, R. Carli "Nonlinear Model Predictive Control of a BMI-guided Wheelchair for Navigation in Unknown Environments" - 2024 *European Control Conference (ECC)*

M. R. ur Rahman, P. Simonetto, A. Polato, F. Pasti, L. Tonin, S. Vascon "OpenNav: Efficient Open Vocabulary 3D Object Detection for Smart Wheelchair Navigation" - *Twelfth International Workshop on Assistive Computer Vision and Robotics (ACVR)* P. Simonetto, S.

Tortora, E. Menegatti, L. Tonin "Fusing user's intention and environment information in BMI driven robotic devices" - *In Proceedings of 5th I-RIM Conference*

Experience:

NEUROROBOTICS University of Padua

02/01/2023 - present, IASLAB at unipd

- Developing a BCI-driven navigation system for wheelchairs, ensuring user safety and comfort.
- Research supervised by L. Tonin.

NEUROROBOTICS UT at Austin, Ph.D. Mobility period

11/01/2024 - 30/04/2025, José del R. Millán lab

- Conducting real-time Error-Related Potential (ErrP) detection for powered wheelchair teleoperation.

WORKSHOP ORGANIZER WS5 at Summer School on Neurorehabilitation

08/06/2025 - 13/06/2025, SSNR

- Organized and managed a workshop on how to connect a BCI to a real device

TEACHING ASSISTANT University of Padua - Intelligence Robotics

01/10/2025 - 28/01/2026 & 02/10/2023 - 02/02/2024

- Manage the course laboratories, where the students learn how to use robots in a dynamic environment through ROS/ROS2 and C++

BCI Cybatlon BCI race - WHI team (2nd Position)

27/10/2024, ETH Zurich

- Designed and optimized MI decoding strategies to enhance BCI performance in competition settings.

ERASMUS UCLouvain University

09/2021 - 01/2022, Louvain-la-Neuve

- Focused on Bioengineering & Neurosciences research.

Internship Reflex&Allen

08/2016 - 09/2016, Wales, UK

- Assisted in company IT infrastructure management and troubleshooting.